DBSSM: A Dual-Branch Spectral–Spatial Mamba Network for Hyperspectral Image Classification

Hanzhong Li, Hua Huang, Hongfeng Li, Jiangtao Peng, *Senior Member, IEEE*, Bengong Zhang, Yazhou Shi, and Huizhen Li

***Abstract*—Hyperspectral image (HSI) classification has been widely applied in numerous fields. Although deep learning-based methods have improved classification performance, existing approaches still struggle to balance accuracy and computational efficiency. Convolutional neural network (CNN)-based methods are limited by local receptive fields, whereas Transformer-based methods suffer from high computational complexity, restricting their application in large-scale scenarios. To achieve collaborative optimization of classification accuracy and computational efficiency, a dual-branch spectral–spatial Mamba (DBSSM) network is proposed, which employs a dual-branch architecture with an interactive feature fusion strategy for efficient joint spectral–spatial modeling. The main contributions are as follows: 1) a spatial dual-scan Mamba (SDSM) module is developed to extract rich sequential features via a dual-directional scanning strategy and shared mechanism, enabling long-range dependency modeling with relatively low computational complexity; 2) a spectral group attention (SGA) module is designed to reduce the number of parameters via a grouping strategy while enhancing spectral information interaction using a self-attention mechanism; and 3) an interactive attention fusion module (IAFM) is proposed to achieve bidirectional interactive fusion between spatial and spectral features, thereby further enhancing the feature learning capability. Experimental results on four benchmark hyperspectral datasets demonstrate that DBSSM outperforms state-of-the-art methods in classification performance while maintaining a low parameter count and computational cost, validating its effectiveness. The code is available at https://github.com/Present-Li/DBSSM**

***Index Terms*—Attention mechanism, hyperspectral image (HSI) classification, information interaction, Mamba.**

Received 12 May 2026; revised 1 July 2026; accepted 4 August 2026. Date of publication 7 August 2026; date of current version 18 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 12401669 and Grant 12371480, in part by Wuhan Textile University Special Fund under Grant 2025438, and in part by Wuhan College Research Fund under Grant JJB202609. *(Hanzhong Li, Hua Huang, and Hongfeng Li are co-first authors.) (Corresponding author: Huizhen Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3721475

I. Introduction

HYPERSPECTRAL images (HSIs) are acquired by imaging spectrometers over dozens to hundreds of contiguous spectral bands [1], [2]. The abundant spectral information enables HSIs to effectively characterize subtle spectral differences among various ground materials [3]. Owing to their advantages, HSIs have been widely applied in numerous fields, such as agricultural monitoring [4], [5], environmental assessment [6], [7], resource investigation [8], and urban planning [9].

Early HSI classification methods were mainly based on traditional machine learning and statistical learning techniques, such as support vector machines (SVMs) [10], random forests (RFs) [11], multinomial logistic regression (MLR) [12], and maximum likelihood classification (MLC) [13]. These methods typically take pixel-wise spectral features as the primary input and construct discriminative models in predefined feature spaces [14], [15]. However, they heavily rely on handcrafted feature design and struggle to effectively model complex nonlinear spectral–spatial relationships. Moreover, their performance is highly sensitive to feature representation and model assumptions, resulting in limited generalization ability and poor robustness, particularly in sample-scarce or complex scenarios.

Deep learning-based HSI classification methods have become a mainstream research direction due to their end-to-end feature learning capability, and various representative architectures have been proposed, such as stacked autoencoders (SAEs) [16], convolutional neural networks (CNNs) [17], generative adversarial networks (GANs) [18], and vision Transformers (ViTs) [19]. Among them, CNNs and their variants play a dominant role in HSI classification owing to their local connectivity and weight-sharing mechanisms, which enable effective modeling of local spectral–spatial correlations [20], [21]. For instance, Roy et al. [22] proposed a hybrid spectral CNN (HybridSN), which integrates 3-D and 2-D convolutions for joint spectral–spatial feature extraction. Liu et al. [23] proposed a spatial–spectral enhancement and fusion network (SSEFN), which improves feature discriminability via a spatial–spectral enhancement strategy and alleviates overfitting through adaptive decision fusion. Lu et al. [24] proposed a separable deep graph convolutional network integrated with CNN and prototype learning (SDGCP), which models interpixel structural relationships

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and enhances interclass separability. Despite the performance improvements achieved by the aforementioned methods, they still struggle to effectively highlight discriminative features, while their capability to suppress noise and interference from heterogeneous pixels remains limited.

To focus on discriminative features and suppress noise and interference, attention mechanisms have been introduced into deep neural networks to adaptively reweight feature representations, thereby enhancing model expressiveness and overall performance [25], [26]. For instance, Hang et al. [27] proposed the spectral spatial attention (SSAtt) network, which embeds spectral and spatial attention modules into CNNs, where adaptive reweighting is employed to emphasize discriminative feature representations and improve feature extraction capability. Fu et al. [28] proposed an attention-enhanced resblock-in-resblock network with an equiangular tight frame classifier, which strengthens feature representation through channel attention and employs a fixed classifier to improve classification performance under limited labeled samples. Sun et al. [29] proposed a dual-branch network named spatial–spectral-channel 3-D channel attention mechanism (3DAM-CAM) DenseNet, which jointly models the complex correlations among spectral, spatial, and channel dimensions, thereby effectively enhancing the discriminability and robustness of the learned feature representations. Fu et al. [30] proposed an attention-enhancing residual and spatial-channel attention-based network (ReSC-net), which combines channel attention-enhanced residual learning with spatial-channel attention to improve spatial–spectral feature representation. Although CNN-based HSI classification methods have improved classification performance to some extent, their limited receptive fields restrict the effective modeling of long-range dependencies.

To alleviate this limitation, the Transformer has been introduced and widely applied in hyperspectral remote sensing analysis. For instance, Qiu et al. [31] proposed a cross-channel dynamic spatial–spectral fusion Transformer (CDSFT), which adaptively integrates cross-channel information to collaboratively model spectral continuity and spatial context. Jiang et al. [32] proposed a graph generative structure-aware Transformer (GraphGST), which incorporates a graph generation-based structural modeling mechanism into the Transformer framework to enhance the representation of interpixel structural relationships. Chen et al. [33] proposed a spectral query spatial Transformer (SQSFormer), where the central pixel guides the spatial self-attention computation to improve feature discriminability. Xu et al. [34] proposed a dual selective fusion Transformer network (DSFormer), which learns optimal spatial–spectral receptive fields through kernel selective fusion and adaptively selects informative tokens during self-attention fusion to improve HSI classification performance. The aforementioned methods enhance the global dependency modeling capability of the network by introducing Transformer architectures, thereby effectively improving HSI classification performance.

The powerful learning capability of Transformers has enabled their widespread adoption and further development in the evolution of foundation models. In particular, the powerful data processing capability of foundation models, combined with their exceptional representation learning ability, has achieved remarkable success in various tasks. This has driven the paradigm of HSI classification from task-specific deep networks toward large-scale remote sensing foundation models. For instance, HyperSIGMA [35] constructs a large-scale hyperspectral pretraining dataset and introduces a sparse sampling attention mechanism, developing a billion-parameter foundation model for HSI analysis. HyperFree [36] incorporates a learnable wavelength-weight dictionary, enabling channel-adaptive hyperspectral representation learning and tuning-free cross-task generalization. Benefiting from large-scale pretraining datasets and self-supervised representation learning, these foundation models have exhibited remarkable generalization capability across diverse downstream tasks, emerging as a promising research direction for HSI analysis.

Although Transformer-based methods have significantly improved HSI classification performance, the quadratic computational complexity of the self-attention mechanism with respect to the input sequence length results in substantial computational and memory overhead, leading to efficiency bottlenecks when processing high-dimensional hyperspectral data [37]. To address this limitation, a selective structured state space model (SSM) named Mamba [38] has emerged as an efficient alternative with linear computational complexity and has attracted considerable attention in remote sensing image analysis. Yang et al. [39] proposed the HSIMamba, which captures long-range spectral dependencies through bidirectional state modeling along the spectral sequence. Huang et al. [40] extended Mamba to joint spectral–spatial modeling for efficient global representation of spectral–spatial features. Li et al. [41] introduced a pyramidal spectral attention mechanism and developed the HyPyraMamba, improving multiscale feature modeling capability and robustness. He et al. [42] proposed interval group spatial–spectral Mamba (IGroupSS-Mamba), which utilizes interval spectral grouping and directional spatial–spectral Mamba scanning to efficiently model multidirectional global contextual dependencies in HSI. Wang et al. [43] proposed spatial–spectral SSM (S^2Mamba), which employs separate spatial and spectral Mamba branches to capture long-range dependencies along spatial and spectral dimensions, followed by feature fusion for HSI classification. Mamba enables efficient processing of high-dimensional long-sequence data with linear complexity, providing a promising solution for large-scale HSI modeling [44].

The aforementioned Mamba-based methods alleviate the high computational cost incurred by Transformers through linear computational complexity. However, some of these methods rely on a single-branch architecture for feature modeling, which may not sufficiently capture the complex spectral–spatial dependencies inherent in hyperspectral data, thereby limiting the model's feature representation capability. Furthermore, although other approaches adopt a dual-branch architecture to model spectral and spatial information, they either apply equal weights to both branches for feature enhancement and fusion, or employ mutual suppression strategies for weighted integration. Such mechanisms may lead to the loss of subtle discriminative information or weaken

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the synergistic representation of spectral and spatial features. Therefore, how to fully leverage the strengths of advanced architectures such as Mamba and Transformer to construct a more effective feature extraction framework, as well as how to design efficient feature enhancement and fusion mechanisms, still requires further investigation.

To improve the collaborative optimization between classification performance and computational efficiency, this article proposes a dual-branch spectral–spatial Mamba (DBSSM) network to fully exploit the spectral and spatial information for each pixel. Considering the computational overhead introduced by high-dimensional feature representations in spatial feature modeling, a spatial dual-scan Mamba (SDSM) module is designed based on the Mamba framework. The SDSM effectively captures rich spatial sequential dependencies through a dual-directional scanning strategy, while leveraging a shared Mamba structure to enhance global spatial modeling capability with low computational complexity. Given the strong global modeling capability of self-attention in Transformers and the fact that the spectral branch only uses the central pixel's spectral information, resulting in a much lower feature dimension compared with the spatial branch, we propose a spectral group attention (SGA) module to model global correlations across spectral bands. In addition, a grouping strategy is introduced to further reduce the feature dimensionality, enabling the model to exploit the global modeling capability of self-attention more effectively while improving computational efficiency. After obtaining the spatial and spectral features, an interactive attention fusion module (IAFM) is designed to further achieve adaptive enhancement and fusion of these features. The IAFM adaptively enhances branch-specific features through mutual guidance between spatial and spectral representations and performs HSI classification based on the fused enhanced features. The proposed IAFM not only strengthens cross-branch information interaction but also significantly promotes the collaborative representation and discriminative capability of spectral–spatial features.

The main contributions of this article are summarized as follows.

1) A novel DBSSM network is proposed, which separately performs feature learning via spatial and spectral branches. The proposed framework achieves effective feature representation while maintaining a favorable tradeoff between classification performance and computational complexity.
2) The constructed SDSM captures rich sequential information through a dual-directional scanning strategy and employs a shared Mamba mechanism to efficiently model spatial dependencies. The designed SGA reduces feature dimensionality and model complexity through a grouping strategy while effectively enhancing spectral features based on the self-attention mechanism.
3) An IAFM is constructed to enable bidirectional interaction and adaptive fusion between spectral and spatial branches, further enhancing the interactive capabilities of the network. Extensive experiments on four benchmark datasets validate the effectiveness of the proposed method.

The remaining sections of this article are organized as follows. Section II describes the details of the proposed network. In Section III, a series of experiments and analyses are illustrated to compare the proposed model with several state-of-the-art methods. Finally, Section IV offers a brief conclusion.

II. Proposed Network

A. Overview

In this section, the proposed DBSSM network is described in detail, and its overall framework is illustrated in Fig. 1. Let $\mathrm{X} \in \mathbb{R}^{H\times W\times C}$ denote the original HSI, where H and W represent the spatial dimensions, and C denotes the number of spectral bands. As shown in Fig. 1, the DBSSM constructs a spatial branch and a spectral branch to model HSI features, and the designed IAFM is employed to achieve the interaction and fusion of features. In the spatial branch, the input X is first processed by PCA [45] to reduce the spectral dimensionality, and local image patches centered at each pixel are extracted. Subsequently, convolution, batch normalization, and the designed SDSM module are applied to progressively learn spatial feature representations. In the spectral branch, the spectral vector of the central pixel is grouped into three nonoverlapping subsets, followed by convolution, batch normalization, and a self-attention mechanism to model and enhance spectral features, thereby effectively capturing the dependencies among different spectral bands. To strengthen the feature interaction capability of the network, the outputs of the two branches are fed into the designed IAFM module, which enables bidirectional interaction and fusion between spatial and spectral features. Finally, the fused spectral–spatial features are passed into a classifier to obtain the final classification results. In Sections II-B–II-D, each component of the proposed DBSSM will be described in detail.

B. Spatial Branch

Considering the high dimensionality and redundancy characteristics of HSIs, PCA is first applied to the original input HSI data X along the spectral dimension to reduce dimensionality, resulting in a reduced image $\mathrm{X}' \in \mathbb{R}^{H\times W\times C'}$. Subsequently, local image patches $\mathrm{F} \in \mathbb{R}^{S\times S\times C'}$ centered at each pixel are extracted, where S represents the spatial size of the image patch and C' denotes the number of spectral bands after dimensionality reduction. To enhance the representation capability of the input features, convolution, batch normalization, and nonlinear mapping operations are sequentially applied to F along the spatial direction to produce the feature map $\mathrm{F}_1 \in \mathbb{R}^{S\times S\times d}$. This process can be represented as follows:

$$\mathrm{F}_1 = \sigma_1\left(BN\left(Conv2D_{1\times 1}\left(\mathrm{F}\right)\right)\right) \tag{1}$$

where $Conv2D_{1\times 1}(\cdot)$ denotes a 2-D convolution operation with kernel size of 1, $BN(\cdot)$ represents batch normalization, and $\sigma_1(\cdot)$ is the ReLU activation function.

After obtaining the preliminarily processed feature F_1, it is fed into the designed SDSM module, whose structure is illustrated in Fig. 2. To fully model long-range spatial contextual information in land-cover distributions, spatial features

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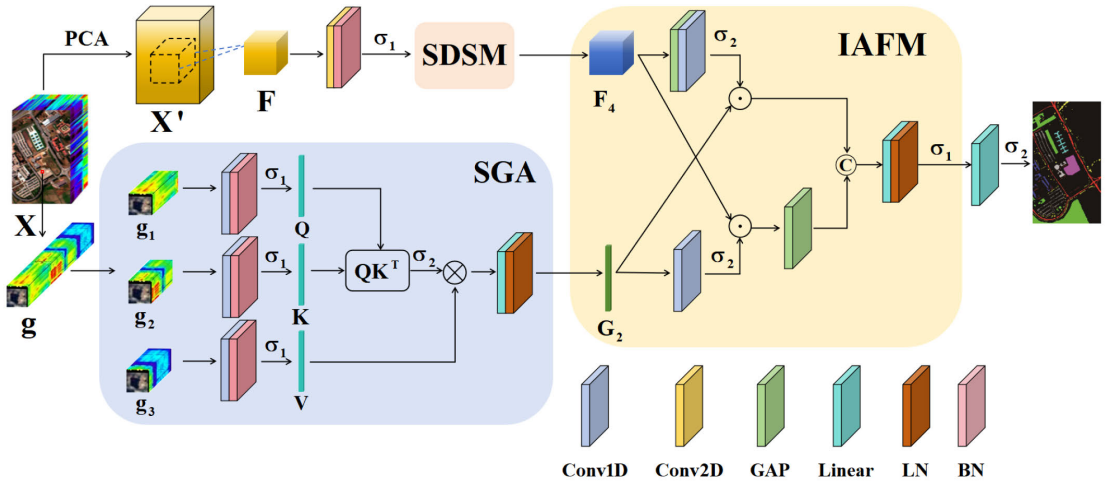


Fig. 1. Overall architecture of the proposed DBSSM, which mainly consists of an SDSM, an SGA, and an IAFM.

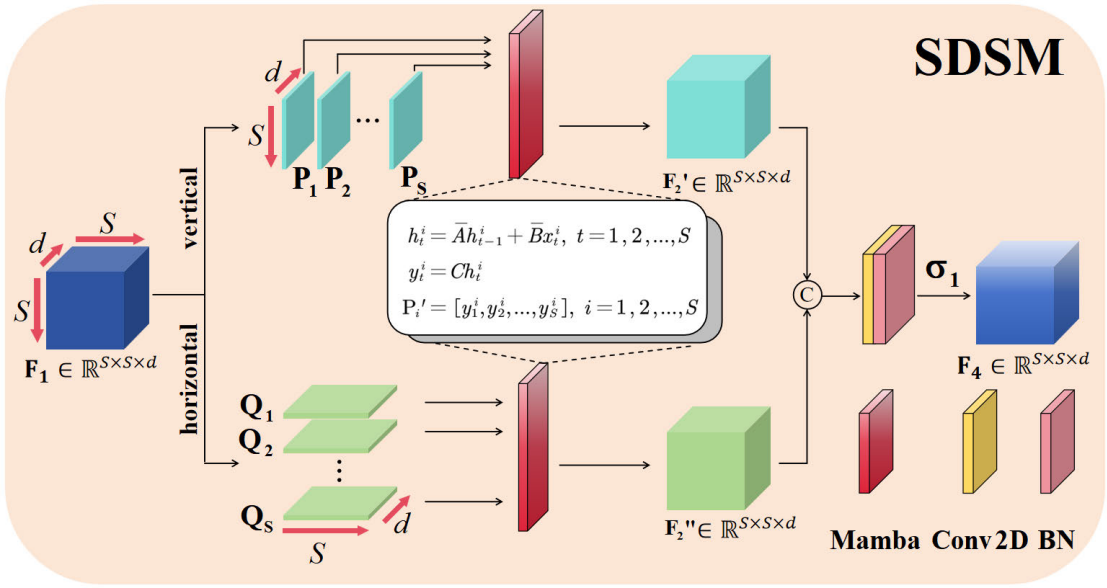


Fig. 2. Detailed structure of SDSM.

are sequentially scanned along both the vertical and horizontal directions, respectively. This dual-directional scanning strategy not only captures rich spatial sequential dependencies but also enables efficient feature learning with a reduced number of parameters through subsequent parallel computation and parameter-sharing mechanisms. In particular, F_1 is first scanned along the vertical and horizontal directions, respectively, resulting in a set of vertical sequences $P_i \in \mathbb{R}^{S\times d}(i = 1,2,\ldots,S)$ and a set of horizontal sequences $Q_j \in \mathbb{R}^{S\times d}(j = 1,2,\ldots,S)$. Subsequently, the sequence sets P_i and Q_j are separately fed into a shared Mamba module for parallel processing, yielding the corresponding feature representations $F_2' \in \mathbb{R}^{S\times S\times d}$ and $F_2'' \in \mathbb{R}^{S\times S\times d}$. The detailed implementation of this process is described as follows.

For the sequence set obtained by vertical scanning, each sequence can be represented as follows:

$$P_i = \left[x_1^i, x_2^i, \ldots, x_S^i\right], \quad i = 1,2,\ldots,S \tag{2}$$

where x_t^i denotes the input feature at the tth position of sequence P_i. We first perform state updates for each sequence P_i, which can be formulated as follows:

$$h_t^i = Ah_{t-1}^i + Bx_t^i, \quad t = 1,2,\ldots,S \tag{3}$$

$$y_t^i = Ch_t^i \tag{4}$$

where h_t^i represents the tth hidden state of sequence P_i and y_t^i denotes the corresponding output feature. $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times L}$, and $C \in \mathbb{R}^{L\times N}$ denote the state transition matrix, input projection matrix, and output projection matrix, respectively. Since hyperspectral data are typically represented as discrete sequences, Mamba adopts the zero-order hold (ZOH) technique [46] to convert the continuous parameters A and B into their discrete counterparts $\bar{A}$ and $\bar{B}$. This process can be expressed as follows:

$$\bar{A} = \exp(\Delta A) \tag{5}$$

$$\bar{B} = (\Delta A)^{-1}\left(\exp(\Delta A) - I\right) \cdot \Delta B. \tag{6}$$

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After discretization, the state update process for each scanned sequence P_i can be formulated as follows:

$$h_t^i = \bar{A}h_{t-1}^i + \bar{B}x_t^i, \quad t = 1, 2, \ldots, S \tag{7}$$

$$y_t^i = Ch_t^i. \tag{8}$$

Then, we can obtain the output sequences

$$P_i' = \left[y_1^i, y_2^i, \ldots, y_S^i\right], \quad i = 1, 2, \ldots, S. \tag{9}$$

After obtaining the output sequences P_i', all sequences are rearranged according to their original spatial positions, and the vertically enhanced feature cube F_2' is reconstructed as follows:

$$F_2' = Concat\left(P_1', \ldots, P_S'\right) \tag{10}$$

where $Concat(\cdot,\cdot)$ represents the concatenation operation.

For the horizontally scanned sequences Q_j, the same state update and rearrangement process as for the sequences P_i is performed to obtain the horizontal feature cube F_2''.

Next, the obtained F_2' and F_2'' are concatenated to form the multidirectional feature $F_3 \in \mathbb{R}^{S\times S\times 2d}$. The feature F_3 is further processed through a convolution layer, batch normalization, and a nonlinear activation function in sequence to refine and fuse the features, yielding the deeper feature representation $F_4 \in \mathbb{R}^{S\times S\times d}$. This procedure can be formulated as follows:

$$F_3 = Concat\left(F_2', F_2''\right) \tag{11}$$

$$F_4 = \sigma_1\left(BN\left(Conv2D_{1\times 1}\left(F_3\right)\right)\right). \tag{12}$$

In the SDSM module, the dual-directional scanning strategy performs sequential modeling along two spatial directions, effectively capturing spatial contextual information from different orientations. Compared with unidirectional scanning, it captures richer spatial information, while compared with multidirectional scanning, it reduces redundant information and computational overhead, thereby achieving a better balance between feature representation and computational efficiency. Meanwhile, the Mamba module, based on the SSM, is capable of modeling long-range dependencies while maintaining linear computational complexity. In addition, by performing parallel processing of sequences from different directions and employing parameter sharing within each direction, the model reduces parameter redundancy and further improves both training and inference efficiency.

C. Spectral Branch

The spectral branch achieves spectral information extraction and enhancement through the constructed SGA module, as illustrated in the blue region of Fig. 1. To alleviate the parameter burden of the self-attention mechanism, a grouping strategy is first adopted to divide the input spectral vector $g \in \mathbb{R}^C$ into three nonoverlapping subvectors $g_i \in \mathbb{R}^D, i = 1, 2, 3$. Excessive grouping may significantly weaken information interaction across different groups, thereby limiting the modeling of global spectral dependencies. Therefore, dividing the spectral features into three groups not only preserves spectral information through low-dimensional representations but also facilitates the construction of intermediate representations corresponding to queries, keys, and values for subsequent cross-group self-attention modeling. After grouping, each subvector g_i is individually processed through convolution, batch normalization, and a nonlinear mapping operation to generate the corresponding query $Q \in \mathbb{R}^{d\times 1}$, key $K \in \mathbb{R}^{d\times 1}$, and value $V \in \mathbb{R}^{d\times 1}$, which can be formulated as follows:

$$D = \lfloor C/3 \rfloor \tag{13}$$

$$Q = \sigma_1\left(BN\left(Conv1D_{k=3}\left(g_1\right)\right)\right) \tag{14}$$

$$K = \sigma_1\left(BN\left(Conv1D_{k=3}\left(g_2\right)\right)\right) \tag{15}$$

$$V = \sigma_1\left(BN\left(Conv1D_{k=3}\left(g_3\right)\right)\right) \tag{16}$$

where $\lfloor\cdot\rfloor$ denotes the round down operation, $Conv1D_{k=3}(\cdot)$ refers to a 1-D convolution with a kernel size of 3, and d denotes the feature dimension.

After this stage, Q, K, and V are fed into the self-attention mechanism for cross-group interaction modeling, producing the output $G_1 \in \mathbb{R}^{d\times 1}$, which captures the correlations and dependencies among different spectral bands. Furthermore, the output G_1 is sequentially processed by a linear transformation and layer normalization to further refine and enhance the feature representation, thereby yielding a higher level feature representation $G_2 \in \mathbb{R}^{d\times 1}$. The specific calculation process is as follows:

$$W = softmax\left(\frac{QK^T}{\sqrt{d}}\right) \tag{17}$$

$$G_1 = WV \tag{18}$$

$$G_2 = LN\left(Linear\left(G_1\right)\right) \tag{19}$$

where $(\cdot)^T$ denotes the transpose operation, $\sqrt{d}$ denotes the scale factor, $soft\max(\cdot)$ is the normalization function, W represents the attention weight, $Linear(\cdot)$ represents the linear layer, and $LN(\cdot)$ denotes layer normalization.

Here, the grouping strategy effectively reduces the dimensionality of the feature vectors, thereby decreasing both the number of model parameters and computational overhead. Meanwhile, the self-attention mechanism enables cross-group spectral information modeling, which fully captures the interband dependencies and thus significantly enhances the representation capability of spectral features.

D. Interactive Attention Fusion Module

To enhance the feature learning capability of the network, the spatial-branch output F_4 and the spectral-branch output G_2 are fed into the proposed IAFM module to achieve cross-branch feature fusion and information interaction. The architecture of IAFM is illustrated in the yellow region of Fig. 1. Within this module, G_2 is first processed by a convolutional layer and a nonlinear mapping to generate a weight representation $G_3 \in \mathbb{R}^{d\times 1}$, which is subsequently applied to F_4 via element-wise multiplication to obtain the weighted spatial feature $F_5 \in \mathbb{R}^{S\times S\times d}$. This step can be expressed as follows:

$$G_3 = \sigma_2\left(Conv1D_{k=1}\left(G_2\right)\right) \tag{20}$$

$$F_5 = G_3 \odot F_4 \tag{21}$$

where $Conv1D_{k=1}(\cdot)$ represents the 1-D convolution with a kernel size of 1, $\sigma_2(\cdot)$ denotes the softmax activation function, and $\odot$ represents element-wise multiplication.

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE IP, PU, SA, AND LK DATASETS

NO.	IP			PU			SA			LK		
	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	Alfalfa	5	41	Asphalt	66	6565	Brocoli_green_weeds_1	20	1989	corn	172	34339
2	Corn-notill	143	1285	Meadows	186	18463	Brocoli_green_weeds_2	37	3689	cotton	42	8332
3	Corn-mintill	83	747	Gravel	21	2078	Fallow	20	1956	sesame	15	3016
4	Corn	24	213	Trees	31	3033	Fallow_rough_plow	14	1380	broad-leaf soybean	316	62896
5	Grass-pasture	48	435	Metal Sheets	13	1332	Fallow_smooth	27	2651	narrow-leaf soybean	21	4130
6	Grass-trees	73	657	Bare Soil	50	4979	Stubble	39	3920	rice	59	11795
7	Grass-pasture-mowed	3	25	Bitumen	13	1317	Celery	36	3543	water	335	66721
8	Hay-windrowed	48	430	Bricks	37	3645	Grapes_untrained	113	11158	roads and houses	36	7088
9	Oats	2	18	Shadows	10	937	Soil_vinyard_develop	62	6141	mixed weed	26	5203
10	Soybean-notill	97	875				Corn_senesced_green_weeds	33	3245			
11	Soybean-mintill	245	2210				Lettuce_romaine_4wk	11	1057			
12	Soybean-clean	59	534				Lettuce_romaine_5wk	19	1908			
13	Wheat	20	185				Lettuce_romaine_6wk	9	907			
14	Woods	126	1139				Lettuce_romaine_7wk	11	1059			
15	Buildings-Grass-Trees-Drives	39	347				Vinyard_untrained	72	7196			
16	Stone-Steel-Towers	9	84				Vinyard_vertical_trellis	18	1789			
-	Total	1024	9225	Total	427	42349	Total	541	53588	Total	1022	203520

Similarly, the spatial feature F_4 is sequentially processed by global average pooling, convolution, and nonlinear mapping to obtain the weight representation $F_6 \in \mathbb{R}^{d\times 1}$. Then, F_6 is element-wise multiplied with G_2, enabling adaptive spatial-guided enhancement of spectral features, and producing the weighted spectral feature $G_4 \in \mathbb{R}^{d\times 1}$. Its formula is expressed as follows:

$$F_6 = \sigma_2\left(Conv1D_{k=1}\left(GAP\left(F_4\right)\right)\right) \tag{22}$$

$$G_4 = F_6 \odot G_2 \tag{23}$$

where $GAP(\cdot)$ denotes the global average pooling operation.

After obtaining the weighted spatial feature F_5 and spectral feature G_4, global average pooling is first applied to F_5, followed by feature concatenation with G_4 to obtain $H_1 \in \mathbb{R}^{2d\times 1}$. H_1 is then processed through a linear transformation, layer normalization, and a nonlinear mapping to obtain the fused spectral–spatial feature $H_2 \in \mathbb{R}^{d\times 1}$. This process is formulated as follows:

$$H_1 = Concat\left(GAP\left(F_5\right), G_4\right) \tag{24}$$

$$H_2 = \sigma_1\left(LN\left(Linear\left(H_1\right)\right)\right). \tag{25}$$

The IAFM module achieves adaptive feature weighting and fusion through a bidirectional guidance mechanism between the spatial and spectral branches. This cross-interaction paradigm effectively enhances cross-branch information exchange within the network, thereby significantly improving the richness and discriminability of feature representations.

Finally, the obtained spectral–spatial feature H_2 is fed into a linear layer followed by a nonlinear mapping to yield the ultimate classification results. It is formulated as follows:

$$y = \sigma_2\left(Linear\left(H_2\right)\right) \tag{26}$$

where y denotes the class label.

III. EXPERIMENT AND ANALYSIS

A. Dataset Descriptions

To verify the effectiveness of the proposed method, experiments are conducted on four benchmark datasets, including Indian Pines (IP),[1] Pavia University (PU),[2] Salinas (SA),[3] and WHU-Hi-LongKou (LK).[4] The IP dataset has a spatial size of 145 × 145 pixels with a spatial resolution of 20 m, and approximately 200 spectral bands remain after removing water absorption bands. The PU dataset has a spatial size of 610 × 340 pixels with a spatial resolution of 1.3 m, consisting of 103 spectral bands. The SA dataset has a spatial size of 512 × 217 pixels with a spatial resolution of 3.7 m, and approximately 204 spectral bands remain after removing water absorption bands. The LK dataset has a spatial size of 550 × 400 pixels with a spatial resolution of approximately 0.463 m, consisting of 270 spectral bands. Table I presents the class labels as well as the numbers of training and testing samples for the four datasets. The proportions of training samples in the IP, PU, SA, and LK datasets are set to 10%, 1%, 1%, and 0.5%, respectively, and the remaining samples are used for testing.

B. Experimental Settings

To comprehensively evaluate the performance of the proposed method, four representative categories of classification approaches are selected for comparison, including traditional machine learning methods (SVM [10], RF [11]), CNN-based methods (HybridSN [22], SSAtt [27], SSEFN [23]), Transformer-based methods (GraphGST [32], SQSFormer [33]), and Mamba-based methods (SS-Mamba [40], HyPyraMamba [41]). All experiments are implemented on the PyTorch platform and conducted on a single NVIDIA GeForce RTX 4060 Ti GPU with 16 GB of memory. The model parameters are optimized using the Adam optimizer with the learning rate set to 0.001. The number of training epochs is set to 100, and the batch size is 64. The number of retained principal components (PCs) in PCA is set to 30.

To quantitatively evaluate the proposed method, the class-specific accuracy (CA), overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (Kappa) are employed as the evaluation metrics. All methods

[1]IP dataset.
[2]PU dataset.
[3]SA dataset.
[4]LK dataset.

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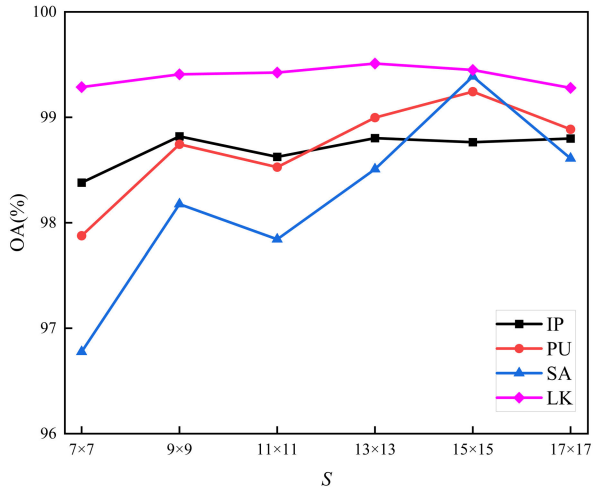


Fig. 3. Impact of different input patch sizes (S) on OA.

TABLE II
IMPACT OF THE NUMBER OF PCS ON ALGORITHM PERFORMANCE

PCs	OA(%)	AA(%)	Kappa(%)	Parameters (M)	FLOPs (G)
15	98.34±0.51	97.88±0.40	98.10±0.58	**0.1082**	**0.0074**
30	**98.82±0.15**	**98.62±0.34**	**98.65±0.17**	0.1092	0.0075
45	98.72±0.32	98.51±0.39	98.54±0.36	0.1102	0.0076
60	98.42±0.98	98.30±0.62	98.20±1.12	0.1111	0.0077

are evaluated using the same training and testing splits to ensure fairness and reliability. The final results are reported as the average of ten independent runs to reduce the bias caused by random sampling.

C. Parameter Analysis

In this section, a series of experiments are conducted to investigate the impact of different parameters on model performance, including the input patch size (S), the number of PCs, and the number of spectral groups (n) in the SGA module.

1) Image Patch Sizes: Here, we present the impact of the input patch size (S) on the performance of the proposed method. The experimental results are shown in Fig. 3. It can be observed that the OA generally increases first and then decreases as S increases, while slight fluctuations can be observed on some datasets. This is mainly because when S is too small, the patch cannot sufficiently cover homogeneous regions, leading to limited contextual information. Conversely, overly large S introduces more heterogeneous pixels, which not only reduce feature discriminability but also significantly increase computational complexity. In addition, due to the large variations in the shape and scale of homogeneous regions in the PU and SA datasets, the curve exhibits slight fluctuations as S increases. Based on the above analysis and experimental results, the patch size S is set to 9, 15, 15, and 13 for the IP, PU, SA, and LK datasets, respectively.

2) Number of PCs: Table II illustrates the impact of the number of PCs on the performance of the proposed algorithm. As observed from Table II, increasing the number of PCs preserves more information, thereby contributing to improved classification accuracy. However, when too many PCs are retained, redundant information is introduced, which not only degrades classification performance but also significantly increases the number of parameters and FLOPs. When the number of PCs is set to 30, the proposed method achieves the highest classification accuracy and the lowest standard deviation while maintaining relatively low computational complexity in terms of parameters and FLOPs. Therefore, we select the top 30 PCs for DBSSM.

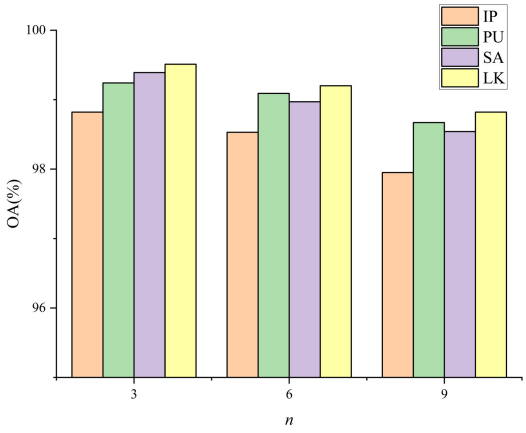


Fig. 4. Impact of different number of groups (n) on OA.

3) Number of Spectral Groups: In the spectral branch, a grouping strategy is adopted to construct a self-attention mechanism for modeling dependencies among spectral bands. Here, we investigate the impact of the number of spectral groups (n) on model performance across four datasets. To generate the corresponding Q, K, and V for self-attention computation, the number of groups is set to be a multiple of 3. Fig. 4 reports the experimental results for n = 3, 6, and 9. As can be observed, the best performance is achieved when n = 3. This is mainly because each group already contains a relatively small number of spectral bands when n = 3. In particular, for the PU dataset, each group consists of 34 bands. When the number of groups becomes excessively large, the interband correlations across different groups cannot be sufficiently modeled, which in turn weakens the global representation capability of spectral information and ultimately degrades the classification performance. Therefore, the number of spectral groups is set to 3 in all experiments.

D. Experiment and Analysis

Tables III–VI present the experimental results on the IP, PU, SA, and LK datasets, where the best results are highlighted in bold. As shown in the tables, the proposed DBSSM method achieves the best performance in terms of OA, AA, Kappa, and most classes across all four datasets. In particular, it demonstrates remarkable performance on small-sample classes, such as classes 1, 7, and 9 in the IP dataset, indicating that the proposed method can effectively alleviate the performance degradation caused by limited training samples and exhibits strong discriminative capability and robustness. In addition, DBSSM consistently yields low standard deviations across different datasets, further validating its stability and generalization ability under varying data splits and complex scenarios.

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TABLE III

CA(%), OA (%), AA(%), KAPPA(%), AND STANDARD DEVIATIONS OF DIFFERENT METHODS ON THE IP DATASET

Class	ML-based		CNN-based			Transformer-based		Mamba-based		
	SVM	RF	HybridSN	SSAtt	SSEFN	GraphGST	SQSFormer	SS-Mamba	HyPyraMamba	DBSSM
1	22.93±7.09	21.71±8.05	97.44±2.56	66.34±10.16	71.50±2.78	**99.76±0.71**	83.90±5.79	95.95±1.35	91.03±1.28	92.68±5.56
2	66.27±1.98	66.86±2.43	96.91±1.44	96.05±2.15	98.11±0.33	95.86±1.18	97.51±0.79	95.36±1.22	**99.18±0.58**	96.78±0.50
3	55.19±4.11	54.24±3.80	90.51±5.95	97.16±1.00	94.97±1.00	94.95±1.53	95.48±1.17	97.06±1.58	96.67±0.21	**99.05±0.78**
4	36.48±3.14	36.43±5.97	86.07±4.48	98.03±1.75	96.84±0.32	96.50±1.93	93.80±2.12	95.79±0.53	95.02±1.49	**99.30±0.57**
5	82.80±3.70	81.29±5.91	92.82±1.34	97.63±1.23	98.52±0.71	94.60±0.88	95.33±0.88	95.85±1.30	97.45±0.61	**98.69±0.71**
6	93.84±1.77	94.22±3.14	94.93±1.05	98.92±0.58	99.70±0.05	99.77±0.17	99.39±0.39	98.20±0.43	99.36±0.16	**99.95±0.10**
7	42.00±12.93	8.40±12.71	39.58±2.08	57.20±11.74	**100.0±0.00**	79.23±5.22	86.00±8.99	**100.0±0.00**	93.75±2.08	**100.0±0.00**
8	96.91±1.45	98.12±0.87	99.14±0.62	99.93±0.11	**100.0±0.00**	**100.0±0.00**	99.91±0.15	99.08±0.65	99.63±0.37	**100.0±0.00**
9	11.67±8.77	7.78±5.67	85.29±2.94	77.22±22.83	89.44±6.78	95.56±5.44	72.78±16.75	71.88±3.12	88.24±5.88	**98.33±2.55**
10	60.61±1.71	67.27±3.26	98.18±0.73	95.85±3.45	**98.97±0.39**	94.98±1.37	98.42±0.69	94.60±3.08	98.73±0.06	98.75±0.31
11	67.27±2.58	86.10±2.51	98.66±0.38	96.52±4.08	99.01±0.18	96.68±0.76	98.88±0.30	97.61±0.61	98.73±0.02	**99.06±0.19**
12	48.15±1.59	45.37±3.28	92.26±4.96	95.02±2.13	98.08±0.84	95.45±0.96	93.31±1.94	99.26±0.11	**99.40±0.00**	97.64±1.44
13	92.86±1.84	92.00±2.60	95.40±1.15	99.95±0.16	**100.0±0.00**	98.97±0.16	**100.0±0.00**	99.70±0.30	99.14±0.29	**100.0±0.00**
14	90.05±1.43	95.40±1.27	99.72±0.19	99.23±0.42	**100.0±0.00**	99.54±0.39	99.93±0.09	98.76±0.25	99.67±0.33	**100.0±0.00**
15	50.06±3.58	39.08±3.99	91.62±7.47	95.59±3.82	97.09±0.90	97.47±2.11	98.67±0.76	95.47±1.29	98.02±1.68	**98.70±0.77**
16	82.74±2.40	77.74±6.86	83.54±0.00	97.62±1.68	**100.0±0.00**	99.05±0.71	98.57±1.28	83.11±14.19	98.10±0.63	98.93±0.99
OA	70.16±0.73	75.23±0.91	96.00±1.74	96.87±0.94	98.46±0.15	96.88±0.18	97.84±0.30	96.97±0.59	98.63±0.14	**98.82±0.15**
AA	62.49±1.28	60.75±1.64	90.13±2.07	91.77±1.94	96.39±0.53	96.15±0.25	94.49±1.50	94.86±0.99	97.01±0.76	**98.62±0.34**
Kappa	65.89±0.80	71.42±1.05	95.43±1.99	96.44±1.06	98.25±0.17	96.44±0.21	97.53±0.34	96.55±0.67	98.44±0.16	**98.65±0.17**

TABLE IV

CA(%), OA (%), AA(%), KAPPA(%), AND STANDARD DEVIATIONS OF DIFFERENT METHODS ON THE PU DATASET

Class	ML-based		CNN-based			Transformer-based		Mamba-based		
	SVM	RF	HybridSN	SSAtt	SSEFN	GraphGST	SQSFormer	SS-Mamba	HyPyraMamba	DBSSM
1	84.61±1.39	84.87±3.63	96.07±1.05	97.36±2.67	**99.71±0.07**	94.60±0.78	95.71±1.89	96.45±0.07	99.14±0.53	99.64±0.22
2	94.40±0.88	95.86±1.11	99.78±0.06	99.88±0.15	99.90±0.02	97.24±3.33	99.85±0.14	**99.96±0.01**	99.77±0.08	99.94±0.04
3	56.66±3.53	42.37±10.41	90.28±1.18	80.86±7.22	87.02±2.31	83.60±2.01	81.26±4.66	87.91±2.91	**97.07±1.70**	90.86±1.20
4	81.69±2.98	78.00±4.21	90.09±3.58	97.54±0.52	**98.58±0.09**	93.15±3.25	97.75±0.40	97.61±0.09	96.88±0.53	97.86±0.14
5	99.74±0.12	98.12±0.56	83.22±0.84	99.97±0.05	**100.0±0.00**	**100.0±0.00**	99.98±0.05	**100.0±0.00**	99.87±0.16	**100.0±0.00**
6	68.77±3.12	39.32±5.17	99.65±0.35	94.61±6.21	99.67±0.06	92.09±4.60	94.73±3.02	99.80±0.14	**99.87±0.21**	99.79±0.03
7	47.26±3.97	64.87±10.12	98.06±1.94	91.39±2.83	99.96±0.04	94.41±0.78	99.26±0.50	96.12±1.11	99.90±0.14	**100.0±0.00**
8	66.04±4.26	81.59±3.80	86.32±2.62	89.73±14.54	97.19±0.31	90.99±2.03	79.86±4.70	93.05±1.14	96.70±1.37	**99.50±0.23**
9	95.30±2.86	99.19±0.25	80.90±2.77	97.53±0.65	99.63±0.07	98.19±0.34	99.18±0.78	98.17±0.17	90.78±1.80	**99.79±0.17**
OA	83.39±0.68	81.56±0.84	95.88±0.75	96.58±1.94	98.88±0.11	94.75±1.20	95.80±0.97	97.89±0.02	98.89±0.18	**99.24±0.12**
AA	77.16±1.05	76.02±1.65	91.60±1.15	94.32±1.80	97.96±0.25	93.81±0.61	94.18±1.26	96.56±0.11	97.78±0.25	**98.60±0.18**
Kappa	77.73±0.92	74.82±1.15	94.53±0.99	95.45±2.59	98.52±0.15	93.05±1.54	94.41±1.29	97.20±0.03	98.52±0.24	**99.00±0.15**

TABLE V

CA(%), OA (%), AA(%), KAPPA(%), AND STANDARD DEVIATIONS OF DIFFERENT METHODS ON THE SA DATASET

Class	ML-based		CNN-based			Transformer-based		Mamba-based		
	SVM	RF	HybridSN	SSAtt	SSEFN	GraphGST	SQSFormer	SS-Mamba	HyPyraMamba	DBSSM
1	98.36±0.94	98.58±0.69	93.83±17.51	**100.0±0.00**	98.08±0.21	99.99±0.02	99.82±0.22	99.76±0.24	**100.0±0.00**	**100.0±0.00**
2	99.45±0.28	99.47±0.19	99.75±0.56	89.61±10.36	99.99±0.01	**100.0±0.00**	98.89±2.16	99.99±0.01	**100.0±0.00**	**100.0±0.00**
3	92.89±2.47	93.84±3.41	92.06±21.94	95.53±1.27	**100.0±0.00**	99.99±0.02	98.57±0.62	99.87±0.13	99.95±0.05	**100.0±0.00**
4	93.06±1.96	98.71±0.68	91.99±19.12	99.96±0.04	99.91±0.03	99.43±0.45	99.37±0.09	**100.0±0.00**	98.22±1.49	99.71±0.29
5	94.64±1.88	98.12±0.49	95.48±7.79	99.28±0.38	98.68±0.06	98.96±0.66	98.73±0.08	**99.96±0.04**	93.27±0.40	99.05±0.43
6	99.71±0.10	99.88±0.07	97.50±7.29	**100.0±0.00**	**100.0±0.00**	99.99±0.02	99.84±0.00	99.97±0.03	99.96±0.04	**100.0±0.00**
7	99.71±0.09	99.54±0.06	99.90±0.12	**100.0±0.00**	99.99±0.01	99.89±0.15	99.03±1.47	99.99±0.01	99.94±0.03	99.99±0.01
8	75.86±3.00	85.79±2.14	98.70±2.74	84.54±5.69	99.11±0.15	94.40±1.52	92.79±2.65	89.02±6.81	**99.70±0.04**	98.73±0.20
9	99.32±1.05	99.70±0.12	99.84±0.26	99.05±0.93	**100.0±0.00**	**100.0±0.00**	99.96±0.08	**100.0±0.00**	**100.0±0.00**	**100.0±0.00**
10	91.90±1.87	87.49±2.59	96.65±3.43	93.12±6.35	97.44±0.07	97.87±0.54	97.84±0.47	**99.89±0.02**	98.38±0.29	98.07±1.02
11	91.81±2.36	90.82±0.58	89.01±14.25	98.81±1.10	97.23±0.36	97.99±0.87	99.39±0.19	99.85±0.05	99.48±0.05	**99.95±0.05**
12	93.25±2.92	98.90±0.69	95.62±9.34	98.62±1.27	99.95±0.00	99.92±0.08	99.98±0.04	**100.0±0.00**	99.82±0.03	99.74±0.25
13	81.59±8.91	90.47±4.55	89.12±29.73	75.95±24.05	**100.0±0.00**	98.45±1.23	98.92±0.62	99.30±0.70	93.94±5.95	99.83±0.29
14	83.06±4.39	90.68±2.29	95.00±9.17	97.52±0.48	97.45±0.19	96.68±2.92	**99.90±0.06**	98.11±0.60	96.46±1.65	98.49±2.40
15	60.85±2.61	48.45±3.02	98.17±2.33	96.79±1.49	90.19±1.14	82.49±3.55	82.86±7.93	98.26±1.51	99.00±0.53	**99.10±0.62**
16	98.07±0.71	93.85±4.08	88.03±24.49	97.77±2.17	99.41±0.06	99.80±0.15	97.76±0.18	99.56±0.32	**99.86±0.14**	99.65±0.39
OA	87.19±0.76	88.08±0.64	97.05±4.72	94.31±2.66	98.08±0.12	96.14±0.65	95.66±0.63	97.39±1.17	99.12±0.21	**99.39±0.11**
AA	90.85±0.61	92.14±0.75	95.04±8.16	95.41±3.02	98.59±0.06	97.87±0.30	97.73±0.32	98.97±0.21	98.62±0.45	**99.52±0.18**
Kappa	85.74±0.85	86.67±0.72	96.71±5.26	93.68±2.95	97.85±0.14	95.70±0.73	95.17±0.71	97.10±1.29	99.03±0.23	**99.32±0.12**

On the contrary, traditional machine learning methods heavily rely on hand-crafted features, which limits their representational capacity and makes it difficult to effectively model the complex spectral–spatial coupling relationships in hyperspectral data. CNN-based methods learn features automatically within local neighborhoods via convolution operations and achieve significant improvements over traditional approaches. However, their global modeling capability

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TABLE VI

CA(%), OA (%), AA(%), KAPPA(%), AND STANDARD DEVIATIONS OF DIFFERENT METHODS ON THE LK DATASET

Class	ML-based		CNN-based			Transformer-based		Mamba-based		
	SVM	RF	HybridSN	SSAtt	SSEFN	GraphGST	SQSFormer	SS-Mamba	HyPyraMamba	DBSSM
1	97.49±0.63	96.55±1.06	99.86±0.06	99.53±0.19	**99.99±0.00**	98.17±1.89	99.70±0.21	99.86±0.02	99.29±0.06	99.68±0.00
2	77.34±3.16	63.48±4.07	81.91±15.63	96.02±1.62	**99.65±0.06**	80.01±10.50	96.00±1.92	97.42±0.22	96.39±0.26	98.09±0.03
3	75.54±3.92	49.65±3.30	95.65±3.48	97.83±0.55	99.52±0.07	96.02±2.59	99.20±0.62	96.36±0.17	95.94±1.67	**99.83±0.07**
4	95.75±0.53	97.25±0.55	99.06±0.53	98.00±1.06	99.24±0.04	97.17±1.52	99.33±0.26	99.53±0.15	98.63±0.29	**99.66±0.02**
5	66.42±6.35	50.29±3.68	84.50±7.81	91.36±3.55	86.96±1.13	91.91±6.30	92.46±2.86	91.54±1.87	92.43±0.59	**95.86±0.99**
6	94.07±2.99	96.60±1.17	99.92±0.08	99.34±0.47	99.41±0.11	94.82±1.75	99.84±0.19	**99.97±0.03**	99.43±0.33	99.80±0.00
7	99.64±0.17	99.90±0.07	99.84±0.11	99.95±0.03	99.93±0.01	99.98±0.01	99.98±0.04	99.98±0.01	99.89±0.10	**100.0±0.00**
8	84.40±3.50	86.72±3.58	70.74±17.53	95.82±0.24	97.18±0.11	94.08±0.91	91.80±6.92	93.79±1.01	88.11±2.94	**98.44±0.33**
9	73.07±1.74	44.65±4.40	89.51±5.02	74.96±1.27	85.99±0.81	83.01±7.29	94.16±2.91	**97.28±0.36**	88.29±3.37	96.22±0.22
OA	94.60±0.33	93.21±0.21	97.22±1.80	98.09±0.22	98.96±0.04	96.83±1.24	98.96±0.36	99.21±0.04	98.31±0.17	**99.51±0.01**
AA	84.86±0.83	76.12±0.69	91.22±5.58	94.76±0.38	96.43±0.18	92.80±2.20	96.94±0.97	97.30±0.12	95.38±0.13	**98.62±0.05**
Kappa	92.88±0.44	90.95±0.28	96.34±2.38	97.49±0.28	98.64±0.05	95.84±1.62	98.64±0.48	98.96±0.05	97.78±0.22	**99.36±0.01**

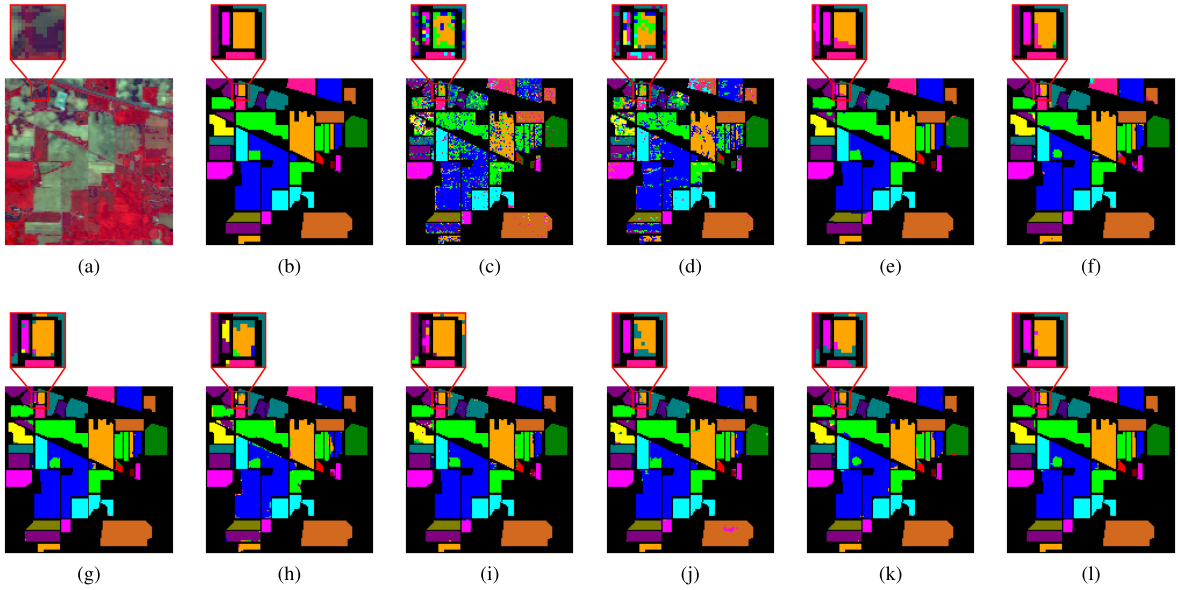


Fig. 5. Classification maps obtained by different methods on the IP dataset. (a) False-color map. (b) Ground truth. (c) SVM. (d) RF. (e) HybridSN. (f) SSAtt. (g) SSEFN. (h) GraphGST. (i) SQSFormer. (j) SS-Mamba. (k) HyPyraMamba. (l) DBSSM.

is still constrained by the limited receptive field. Transformer-based methods enhance long-range dependency modeling through the self-attention mechanism, and they are particularly effective in small-sample scenarios by capturing spatial correlations between pixels and their neighborhoods, leading to notable performance improvements. For instance, on the IP dataset, Transformer-based methods achieve superior classification performance compared with CNN-based methods. Mamba-based methods, benefiting from state-space modeling, not only exhibit strong global modeling capability but also demonstrate better adaptability across different data scenarios. The proposed DBSSM introduces a dual-branch architecture to enhance the interaction and collaborative learning between spectral and spatial features, enabling more effective information fusion and more discriminative feature representations, thereby further improving the overall classification performance.

Figs. 5–8 visually illustrate the classification maps produced by different methods on the four datasets. From the overall comparison, traditional machine learning methods exhibit a considerable number of misclassified pixels, leading to obvious noise and fragmented regions in the classification results. CNN-based methods, constrained by local receptive fields, show certain variations in adaptability across different scenes and still suffer from blurred boundaries and local confusion in complex regions. However, our model effectively reduces misclassified pixels. While maintaining strong spatial consistency, it better preserves land-cover boundaries and achieves consistently satisfactory classification performance across all scenes. These results fully demonstrate that DBSSM exhibits strong stability and good adaptability to different scenarios.

E. t-SNE Visualization

To demonstrate the discriminative capability of the features extracted by DBSSM, t-SNE visualization is conducted on the IP dataset. Representative models are selected from CNN-based, Transformer-based, and Mamba-based approaches for comparison with the proposed method, and the results are shown in Fig. 9. As illustrated in Fig. 9, HybridSN exhibits significant interclass overlap and ambiguous class boundaries. GraphGST substantially improves feature discriminability due to its effective global modeling capability. HyPyraMamba further enhances class separability. The proposed DBSSM

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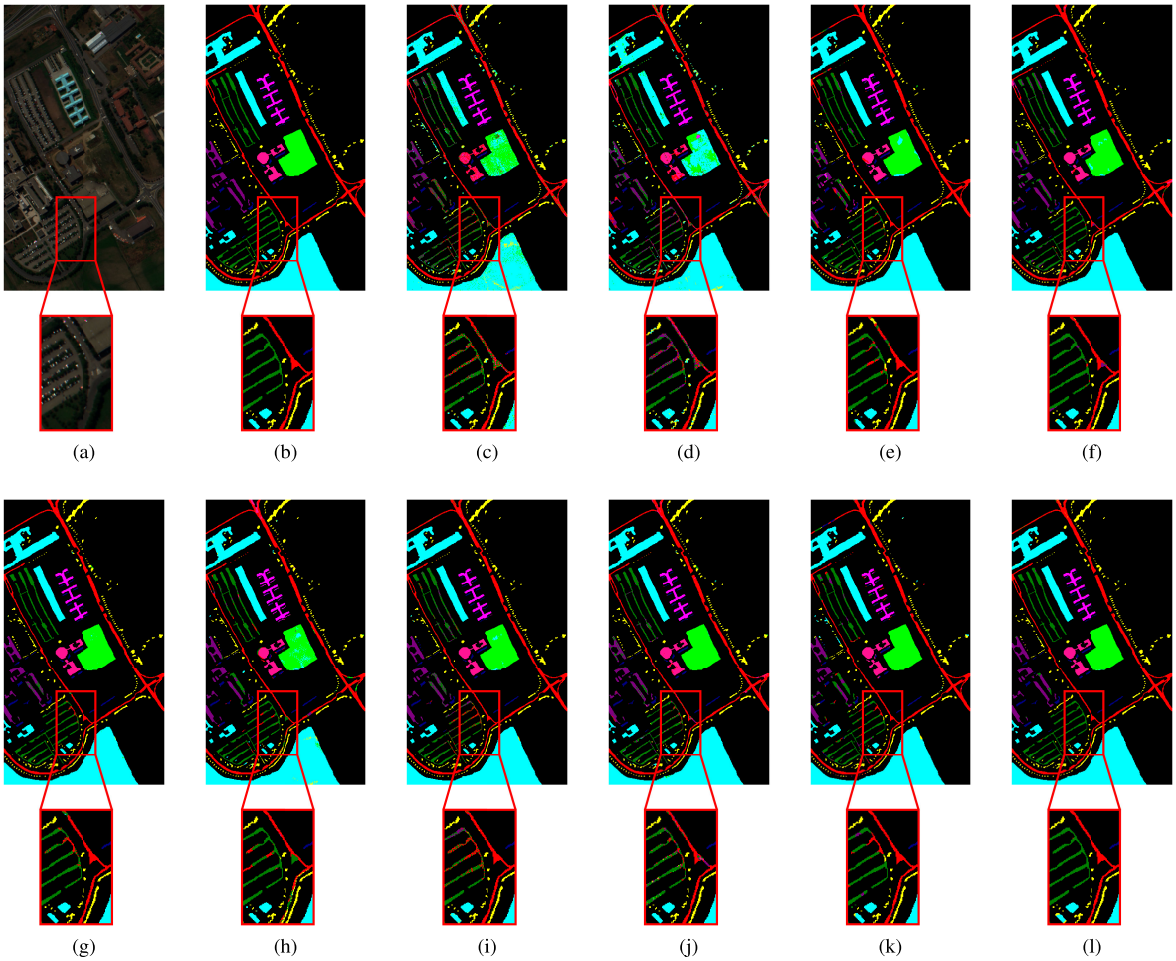


Fig. 6. Classification maps obtained by different methods on the PU dataset. (a) False-color map. (b) Ground truth. (c) SVM. (d) RF. (e) HybridSN. (f) SSAtt. (g) SSEFN. (h) GraphGST. (i) SQSFormer. (j) SS-Mamba. (k) HyPyraMamba. (l) DBSSM.

TABLE VII

EFFECT OF DIFFERENT MODULES ON ALGORITHM PERFORMANCE. "✓" INDICATES THAT THE MODULE IS USED, OTHERWISE IT IS NOT USED

CASE	SDSM	SGA	IAFM	IP			PU			SA			LK		
				OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
1				86.83 ±1.14	80.07 ±1.63	84.98 ±1.29	90.62 ±1.12	84.25 ±2.71	87.57 ±1.48	89.33 ±0.66	87.00 ±1.12	88.12 ±0.74	95.51 ±0.20	86.43 ±1.46	94.09 ±0.27
2	✓			98.50 ±0.21	97.66 ±0.89	98.29 ±0.32	98.79 ±0.45	97.81 ±0.76	98.40 ±0.60	98.56 ±0.72	98.95 ±0.33	98.40 ±0.81	99.19 ±0.06	97.90 ±0.29	98.94 ±0.08
3		✓		98.19 ±0.37	98.18 ±0.75	97.93 ±0.42	98.84 ±0.28	97.93 ±0.55	98.46 ±0.36	97.42 ±0.73	98.25 ±1.00	97.13 ±0.81	99.06 ±0.03	97.03 ±0.46	98.77 ±0.03
4	✓	✓		98.69 ±0.16	98.45 ±0.47	98.50 ±0.19	98.85 ±0.45	98.27 ±0.47	98.47 ±0.59	98.96 ±0.52	99.44 ±0.24	98.84 ±0.58	99.20 ±0.20	98.15 ±0.67	98.95 ±0.26
5	✓	✓	✓	**98.82 ±0.15**	**98.62 ±0.34**	**98.65 ±0.17**	**99.24 ±0.12**	**98.60 ±0.18**	**99.00 ±0.15**	**99.39 ±0.11**	**99.52 ±0.18**	**99.32 ±0.12**	**99.51 ±0.01**	**98.62 ±0.05**	**99.36 ±0.01**

achieves the best performance, presenting more compact intraclass distributions and more distinct interclass separations, which clearly demonstrates its superiority in feature extraction and discriminative representation learning.

F. Ablation Studies

This section investigates the impact of the key components SDSM, SGA, and IAFM within the proposed framework on model performance across four benchmark datasets, as reported in Table VII. When the SDSM, SGA, or IAFM modules are removed, they are replaced by 2-D convolution, 1-D convolution, and feature concatenation, respectively. The configuration without any of these modules (Case 1) is treated as the baseline model. As shown in Table VII, the baseline achieves the lowest performance. Introducing either SDSM (Case 2) or SGA (Case 3) leads to a substantial performance improvement, indicating that both spatial and spectral feature representations play a critical role in classification accuracy. When SDSM and SGA are jointly employed, the model performance is further improved, demonstrating their complementary roles in spatial–spectral feature

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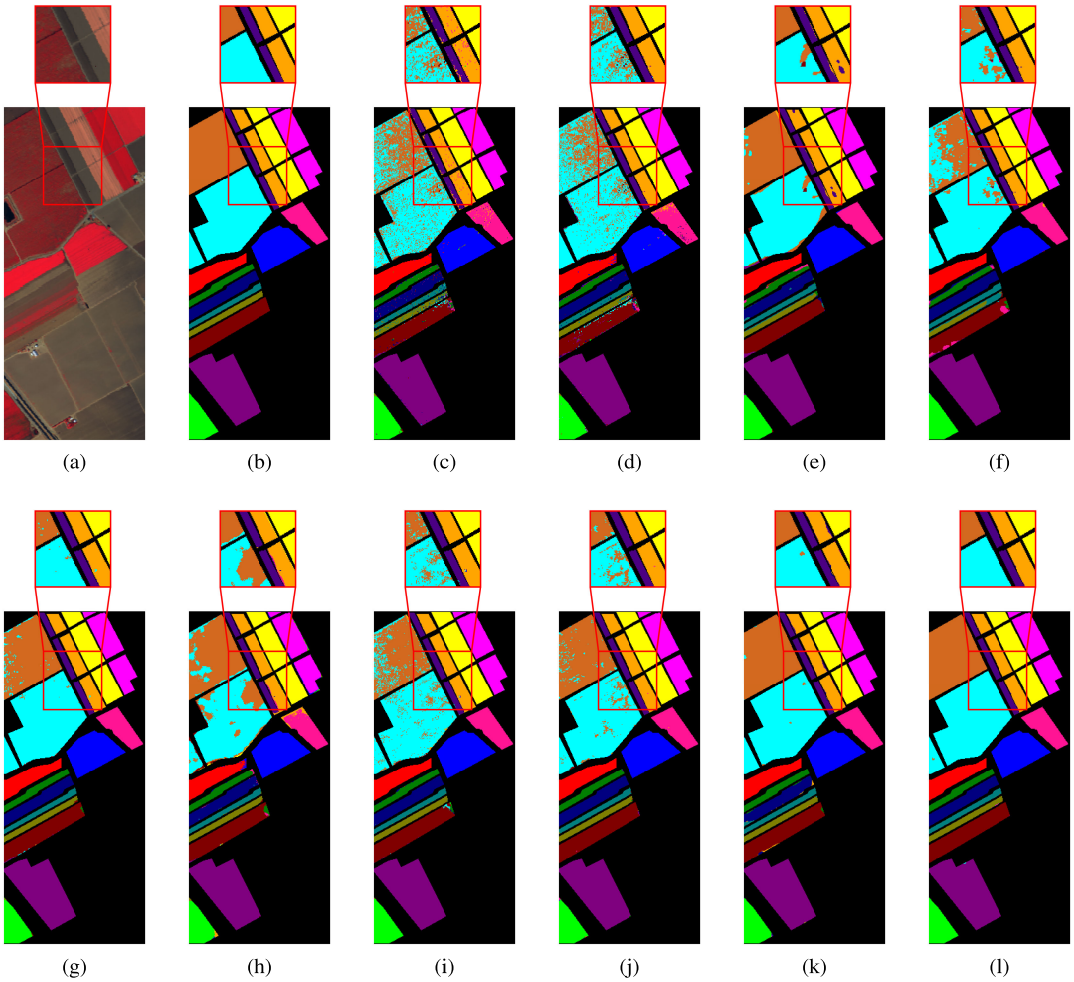


Fig. 7. Classification maps obtained by different methods on the SA dataset. (a) False-color map. (b) Ground truth. (c) SVM. (d) RF. (e) HybridSN. (f) SSAtt. (g) SSEFN. (h) GraphGST. (i) SQSFormer. (j) SS-Mamba. (k) HyPyraMamba. (l) DBSSM.

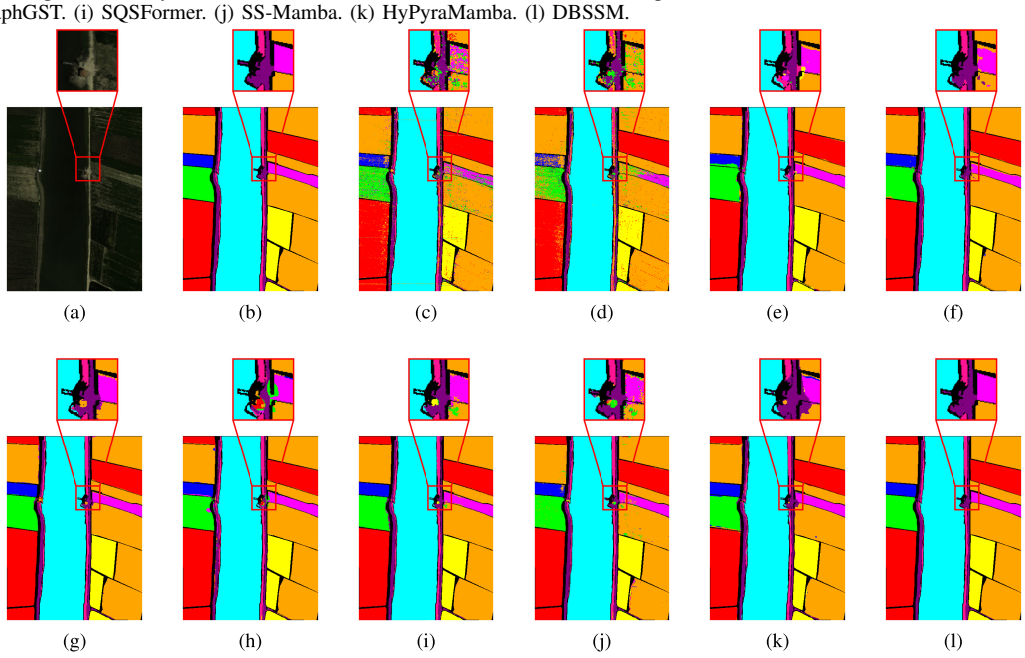
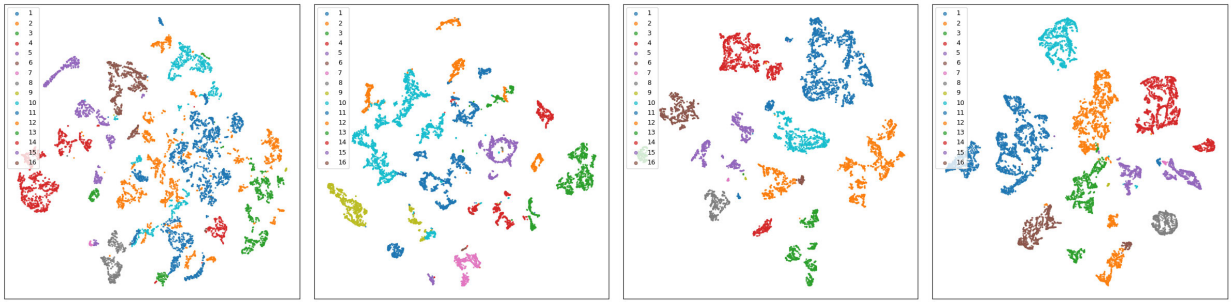


Fig. 8. Classification maps obtained by different methods on the LK dataset. (a) False-color map. (b) Ground truth. (c) SVM. (d) RF. (e) HybridSN. (f) SSAtt. (g) SSEFN. (h) GraphGST. (i) SQSFormer. (j) SS-Mamba. (k) HyPyraMamba. (l) DBSSM.

learning. Finally, the full model integrating all three modules achieves the best performance across all four datasets, yielding the highest classification accuracy and the lowest standard deviation.

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(a) (b) (c) (d)

Fig. 9. T-SNE visualization of different methods on the IP dataset. (a) HybridSN. (b) GraphGST. (c) HyPyraMamba. (d) DBSSM.

TABLE VIII
PERFORMANCE AND COMPUTATIONAL COMPLEXITY ANALYSIS OF DIFFERENT COMPARISON METHODS ON THE IP DATASET

Category	Method	Parameters(M)	FLOPs(G)	Train Time(s)	Test Time(s)	OA(%)	AA(%)	Kappa(%)
CNN-based	HybridSN	5.1222	0.2477	74.26	1.32	96.00±1.74	90.13±2.07	95.43±1.99
	SSAtt	0.5962	0.0225	**17.18**	0.87	96.87±0.94	91.77±1.94	96.44±1.06
	SSEFN	10.2608	2.6748	187.05	5.40	98.46±0.15	96.39±0.53	98.25±0.17
Transformer-based	GraphGST	1.3678	0.1171	21.44	0.82	96.88±0.18	96.15±0.25	96.44±0.21
	SQSFormer	0.3406	0.1122	34.80	1.22	97.84±0.30	94.49±1.50	97.53±0.34
Mamba-based	SS-Mamba	1.0697	0.0091	85.39	2.01	96.97±0.59	94.86±0.99	96.55±0.67
	HyPyraMamba	0.7824	0.0197	90.09	2.03	98.63±0.14	97.01±0.76	98.44±0.16
	DBSSM	**0.1092**	**0.0075**	64.06	**0.79**	**98.82±0.15**	**98.62±0.34**	**98.65±0.17**

G. Complexity Analysis

This section presents a comprehensive comparison of different methods on the IP dataset in terms of classification performance and computational complexity, including the number of model parameters, FLOPs, training and testing time, and classification accuracy, as summarized in Table VIII. Compared with CNN-based methods, Transformer-based methods achieve improvements in both classification performance and computational efficiency. In particular, compared with HybridSN, they significantly reduce the number of parameters, FLOPs, and training time while improving the AA by approximately 5% points and achieving lower standard deviations. Mamba-based methods further achieve linear computational complexity through state-space modeling, which substantially reduces FLOPs while further improving classification performance. Among all compared methods, the proposed DBSSM achieves the lowest number of parameters, FLOPs, and testing time while delivering the best classification performance. Due to its dual-branch architecture and the introduction of a self-attention mechanism in spectral feature modeling, its training time is relatively higher; however, it still outperforms other Mamba-based methods. Overall, DBSSM demonstrates clear advantages in both classification performance and computational efficiency, achieving a favorable tradeoff between classification accuracy and computational complexity.

IV. CONCLUSION

In this article, a novel DBSSM network is proposed, which integrates strong feature modeling capability with lightweight characteristics, achieving an effective tradeoff between classification performance and computational complexity. In the DBSSM, the SDSM module leverages a dual-directional scanning strategy and a shared Mamba architecture to obtain rich spatial contextual information with low computational overhead. In the spectral branch, an SGA module is employed, which combines the grouping strategy with the self-attention mechanism to capture long-range dependencies of spectral sequences while reducing the feature dimensionality. The embedded IAFM enables bidirectional guidance and adaptive fusion between spectral and spatial features, further improving information interaction capability. Experimental results on four public datasets demonstrate that the proposed method not only facilitates the joint optimization of classification performance and computational efficiency but also exhibits strong generalization capability across different scenarios.

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